when GRUB2 starts counting at 0. If this variable is set to the string saved, the entry will be taken from */boot/grub2/grubenv.*

```
GRUB_CMDLINE_LINUX
```

This variable contains a list of extra kernel command-line options that should be added to every single Linux kernel. Typical uses include 'rhgb quiet' for a graphical boot, 'console=xxxxxx' for selecting a kernel console device, and 'crashkernel=xxxx' for configuring automatic kernel crash dumping.

After updating the file */etc/default/grub*, changes can be applied by running the command *grub2-mkconfig –o /boot/grub2/grub.cfg*. This will generate a fresh configuration file using the scripts in */etc/grub.d* and the settings in */etc/defaults/grub*. If no output file is specified with the *–o* option, a configuration file will be written to the standard output.

Reinstalling GRUB2 into the MBR

If, for some reason, the MBR, or the embedding area, has become damaged, an administrator will have to reinstall GRUB2 into the MBR. Since this implies that the system is currently unbootable, this is typically done from some live CD or from within the rescue environment provided by the Anaconda installer.

The following procedure explains how to boot into a rescue environment and reinstall GRUB2 into the MBR from there. If an administrator is still logged into a running system, the procedure can be shortened to just the *grub-install* command.

1. Boot from an installation source; this can be a DVD image, a netboot CD, or from PXE providing a RHEL installation tree.
2. In the boot menu for the installation media, select the 'Rescue an installed system' option, or edit the kernel command line to include the word 'rescue'.
3. When prompted about mounting the disks for the target system to be rescued, select Option 1 (continue). This will mount the system under */mnt/sysimage.*
4. Press *Enter* to obtain a shell when prompted. This shell will live inside the installation/rescue environment, with the target system mounted under */mnt/sysimage*.
   This shell has a number of tools available for rescuing a system, such as all common file systems, disks, LVMs, and networking tools. The various bin directories of the target system are added to the default executable search path *(${PATH})* as well.
5. *chroot* into the */mnt/sysimage* directory, using the following command:

```
# chroot /mnt/sysimage
```

6. Verify that */boot* is mounted, using the command given below. Depending on the type of installation, */boot* can be on a separate partition, or it can be part of the root file system.

```
# ls -l  /boot.
```

7. Use the command *grub2-install* to rewrite the boot loader sections of the MBR and the embedding area. This command will need the block device that represents the main disk as an argument. If unsure about the name of the main disk, use *lsblk* and *blkid* to identify it.

```
# grub2-install /dev/vda
```

8. Reboot the system. This can be done by exiting both the *chroot* shell and the rescue shell.
9. Wait for the system to reboot twice. After the first reboot, the system will perform an extensive SELinux file system relabel, after which it will reboot again automatically to ensure that proper contexts are being used. This relabel is triggered automatically by the Anaconda rescue system creating the */. autorelabel.*

Resolving boot loader issues on UEFI systems

What is UEFI?

Introduced in 2005, the Unified Extensible Firmware Interface (UEFI) replaces the older Extensible Firmware Interface (EFI) and the system BIOS. A replacement for the older BIOS was sought to work around the limits that the BIOS imposed, such as 16-bit processor mode and only 1 MiB of addressable space.

UEFI also allows booting from disks larger than 2 TiB, as long as they are partitioned using a GUID Partition Table (GPT).

One of the big differences between UEFI systems and BIOS systems is the way they boot. While in the case of the latter, the BIOS has to search for bootable devices, operating systems can 'register' themselves with the UEFI firmware. This makes it easier for end users to select which operating system they want to boot in a multi-boot environment — GRUB2 or UEFI.

GRUB2 and UEFI

In order to boot a UEFI system using GRUB2, an EFI system partition (ESP) needs to be present on the disk. This partition should have a GPT UUID or the MBR type. The partition needs to be formatted with a FAT file system, and should be large enough to hold all bootable kernels and the boot loader itself. A size of 512MiB is recommended, but smaller sizes will work. This partition should be mounted at */boot/efi* and will normally be created by the Anaconda installer.